CHAPTER 4

Moving Beyond Linearity

In the previous chapter, we have mostly focused on linear models. Linear models are relatively simple to describe and implement, and have advantages over other approaches in terms of interpretation and inference. However, standard linear regression can have significant limitations in terms of predictive power. This is because the linearity assumption is almost always an approximation, and sometimes a poor one. We have also seen that we can improve upon least squares using ridge regression, the lasso, and Elastic Net. In that setting, the improvement is obtained by reducing the complexity of the linear model, and hence the variance of the estimates. But we are still using a linear model, which can only be improved so far! In this chapter we relax the linearity assumption while still attempting to maintain as much interpretability as possible. We do this by examining very simple extensions of linear models like polynomial regression and step functions, as well as more sophisticated approaches such as splines, local regression, and generalized additive models.

4.1 Polynomial Regression

If we have **nonlinear relationship** between the input variables (features) and the output variable (target), then we may create new features by adding powers of each feature to fit the nonlinear line. Finally, we train a linear model on this augmented set of features. This technique is called Polynomial Regression. The degree of a Polynomial Regression determines the complexity of the model.

The standard way to extend linear regression to settings in which the relationship between the predictors and the response is nonlinear has been to replace the standard linear model.

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

with a polynomial function

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3 + \cdots + \beta_d x_i^d + \epsilon_i,$$

where ε_i is the error term. This approach is known as polynomial regression. For large enough degree d, a polynomial regression allows us to produce an extremely non-linear curve. Notice that the coefficients can be easily estimated using least squares linear regression because this is just a standard linear model with predictors $x_i,\ x_i^2,\ x_i^3, \ldots,\ x_i^d$. Generally speaking, it is unusual to use d greater than 3 or 4 because for large values of d, the polynomial curve can become overly flexible and can take on some very strange shapes.

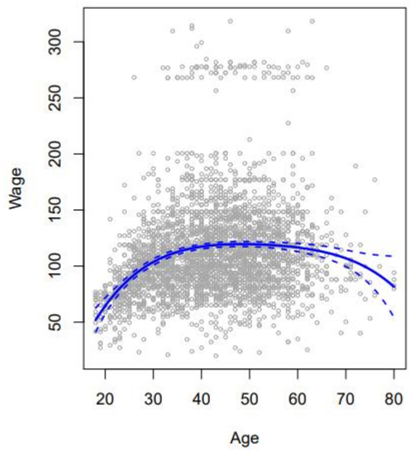

Figure 4.1 The Wage data. The solid blue curve is a degree-4 polynomial of wage (in thousands of dollars) as a function

of age, fit by least squares

Matrix form and calculation of estimates

We have the system of equations:

$$\begin{cases} y_1 = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \beta_3 x_1^3 + \cdots + \beta_d x_1^d + \varepsilon_1 \\ y_2 = \beta_0 + \beta_1 x_2 + \beta_2 x_2^2 + \beta_3 x_2^3 + \cdots + \beta_d x_2^d + \varepsilon_2 \\ y_3 = \beta_0 + \beta_1 x_3 + \beta_2 x_3^2 + \beta_3 x_3^3 + \cdots + \beta_d x_3^d + \varepsilon_3 \\ \vdots \\ y_n = \beta_0 + \beta_1 x_n + \beta_2 x_n^2 + \beta_3 x_n^3 + \cdots + \beta_d x_n^d + \varepsilon_n \end{cases}$$

This system of equations can be written as:

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & x_1 & x_1^2 & x_1^3 & \dots & x_1^d \\ 1 & x_2 & x_2^2 & x_2^3 & \dots & x_2^d \\ 1 & x_3 & x_3^2 & x_3^3 & \dots & x_3^d \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & x_n^3 & \dots & x_n^d \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_d \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

which when using pure matrix notation is written as

$$y = X\beta + \varepsilon$$

The vector of estimated polynomial regression coefficients (using OLS) is

$$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T y$$

General case

In polynomial regression with multiple features, we extend the formulation to include higher-order terms and cross-terms. Let's assume we have mmm features $x_1, x_2, \dots, x_m$ and want to fit a polynomial of degree d. The general form of the model is:

$$y = \beta_0 + \sum_{j=1}^{m} \beta_j x_j + \sum_{j=1}^{m} \sum_{k=j}^{m} \beta_{jk} x_j x_k + \cdots + \beta_d \prod_{j=1}^{m} x_j^{d_j} + \epsilon$$

the terms include all polynomial combinations of the features up to degree d.

Example 1: for 2 Features x_1, x_2 and d=2

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1^2 + \beta_4 x_2^2 + \beta_5 x_1 x_2 + \epsilon$$

The design matrix X is

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & x_{11}^2 & x_{12}^2 & x_{11}x_{12} \\ 1 & x_{21} & x_{22} & x_{21}^2 & x_{22}^2 & x_{21}x_{22} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & x_{n1}^2 & x_{n2}^2 & x_{n1}x_{n2} \end{bmatrix}$$

The coefficient vector

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_5 \end{bmatrix}$$

The output vector

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

Example 2: For a **third-degree polynomial regression** with **three features** (x_1, x_2, x_3), the model includes all possible monomials up to degree 3:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_1^2 + \beta_5 x_2^2 + \beta_6 x_3^2 + \beta_7 x_1 x_2 + \beta_8 x_1 x_3 + \beta_9 x_2 x_3$$

$$+\beta_{10} x_1^3 + \beta_{11} x_2^3 + \beta_{12} x_3^3 + \beta_{13} x_1^2 x_2 + \beta_{14} x_1^2 x_3 + \beta_{15} x_2^2 x_1 + \beta_{16} x_2^2 x_3 + \beta_{17} x_3^2 x_1 + \beta_{18} x_3^2 x_2 + \beta_{19} x_1 x_2 x_3 + \epsilon$$

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & x_{13} & x_{11}^2 & x_{12}^2 & x_{13}^2 & x_{11}x_{12} & x_{11}x_{13} & x_{12}x_{13} & x_{11}^3 & x_{12}^3 & x_{13}^3 & x_{11}^2 x_{12} & x_{11}^2 x_{13} & x_{12}^2 x_{11} & x_{12}^2 x_{13} & x_{13}^2 x_{11} & x_{13}^2 x_{12} & x_{11}x_{12}x_{13} \\ 1 & x_{21} & x_{22} & x_{23} & x_{21}^2 & x_{22}^2 & x_{23}^2 & x_{21}x_{22} & x_{21}x_{23} & x_{22}x_{23} & x_{21}^3 & x_{22}^3 & x_{23}^3 & x_{21}^2 x_{22} & x_{21}^2 x_{23} & x_{22}^2 x_{21} & x_{22}^2 x_{23} & x_{23}^2 x_{21} & x_{23}^2 x_{22} & x_{21}x_{22}x_{23} \\ 1 & x_{31} & x_{32} & x_{33} & x_{31}^2 & x_{32}^2 & x_{33}^2 & x_{31}x_{32} & x_{31}x_{33} & x_{32}x_{33} & x_{31}^3 & x_{32}^3 & x_{33}^3 & x_{31}^2 x_{32} & x_{31}^2 x_{33} & x_{32}^2 x_{31} & x_{32}^2 x_{33} & x_{33}^2 x_{31} & x_{33}^2 x_{32} & x_{31}x_{32}x_{33} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & x_{n3} & x_{n1}^2 & x_{n2}^2 & x_{n3}^2 & x_{n1}x_{n2} & x_{n1}x_{n3} & x_{n2}x_{n3} & x_{n1}^3 & x_{n2}^3 & x_{n3}^3 & x_{n1}^2 x_{n2} & x_{n1}^2 x_{n3} & x_{n2}^2 x_1 & x_{n2}^2 x_{n3} & x_{n3}^2 x_{n1} & x_{n3}^2 x_{n2} & x_{n1}x_{n2}x_{n3} \end{bmatrix}$$

The **coefficient vector** is:

$$\boldsymbol{\beta}^T = \begin{bmatrix} \beta_0 & \beta_1 & \beta_2 & \beta_3 & \beta_4 & \beta_5 & \beta_6 & \beta_7 & \beta_8 & \beta_9 & \beta_{10} & \beta_{11} & \beta_{12} & \beta_{13} & \beta_{14} & \beta_{15} & \beta_{16} & \beta_{17} & \beta_{18} & \beta_{19} \end{bmatrix}^T$$

The number of polynomial terms:

$$p = \binom{m+d}{d} = \binom{3+3}{3} = \binom{6}{3} = \frac{6!}{3!\,(6-3)!} = 20$$